

The Metric Stack of Concern: From Viability Prediction to Maintained Self/World Boundaries in Minimal Agents

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Abstract

How can we tell whether a minimal agent's behavior depends on the intended internal structure, rather than on a proxy? We address this measurement problem in homeostatic bandits where agents must predict viability change, act, intervene through costly null actions, and attribute observed changes to self versus world. We synthesize a 25-study experimental arc in which a *family of minimal homeostatic agents* is progressively augmented with mechanisms for autonomous self/world attribution: concern-like representation, vector-valued valence, active null-anchored intervention (Brehmer et al., 2022), calibrated probe selection (Houlsby et al., 2011; Settles, 2009), decision-layer habituation, and learned probe abstractions. The arc reaches a clearly identified architectural ceiling: in the tested shared-head / null-intervention setup, probe-policy improvements no longer close the role-specific mediated-identifiability gap.

We study **minimal computational precursors of concern-like agency**. We do not claim consciousness, full agency, or general intelligence. The contribution is threefold:

1. The **Metric Stack of Concern** — a 20-layer diagnostic stack (twelve core quantitative metrics expanded into the historical sequence in which they were added) that makes the philosophical thesis "meaning is regulated concern" empirically tractable.
2. The **Correction Chain** — eight named distinctions the program forced (behavior $\neq$ representation, residual scale $\neq$ systematic error, current error $\neq$ value of probing, etc.).
3. A **Positive Mechanism** — a working detect–allocate–saturate–re-engage cycle (Friston, 2010; Dewey, 1938) with three-head world decomposition and learned probe abstractions, plus a precise architectural ceiling at the role-specific mediated identification limit.

The autonomous-probing arc assumes a privileged null intervention; learning or inventing identifying interventions is left for Phase 2. The arc reaches its natural endpoint at the architectural ceiling; further closure requires different research questions (disjoint per-role representations, richer intervention types, multi-agent or continuous-state environments).

This manuscript is self-contained. A reader should be able to reproduce, review, critique, and analyze all key findings without reading any of the 25 prior internal studies. Detailed methodology, per-experiment results, anti-cheat gates, alternative-explanations red-team, falsification conditions, and reproducibility recipes are included. Code, sweep manifests, and figure-regeneration scripts are described in Appendix D and will be released alongside the manuscript.

1. Introduction

The starting thesis is conceptual: meaning is not merely compression or passive latent geometry. Meaning-like structure appears when differences become salient under concern, and agency-like structure appears when that concern is coupled to action, self-maintenance, boundary preservation, repair, and time. This claim is shared across multiple traditions — Heidegger's care-laden world (Heidegger, 1927), Gibson's affordances (Gibson, 1979), Uexküll's *Umwelt* (Uexküll, 1934), the enactive/autopoietic tradition (Maturana & Varela, 1980; Thompson, 2007; Di Paolo, 2005), Ashby's cybernetics (Ashby, 1952), Friston's active inference (Friston, 2010; Parr et al., 2022), Jonas's organism-as-self-concern (Jonas, 1966), Vervaeke's

relevance realization (Vervaeke, 2019), and Simondon's individuation (Simondon, 1958). None of these traditions predicted the specific mechanisms we found, but all predicted the *shape*.

We **do not claim** to have built consciousness, full agency, or general intelligence. We study **minimal computational precursors of concern-like agency**. The contribution is computational and methodological — a working mechanism stack and a metric ladder that makes the philosophical thesis empirically tractable, with sharp boundary conditions.

Scope and key caveats up front. The arc operates in a two-variable homeostatic bandit (energy E , damage D) with four item roles, three actions, and a hand-coded null action whose dynamics the agent uses as a privileged identifying intervention. Both the viability variables and the null action are simulator-defined; learning either the viability dimensions or the intervention type is out of scope. All results are reported across three seeds. Pre-registration discipline was added at Paper 17A; earlier metric-stack layers should be treated as exploratory. Section 16 enumerates these and other limitations; Section 17 lists six falsification conditions under which the maintained-boundary interpretation would weaken.

Why this is a measurement-stack paper, not a benchmark paper. Several reliable patterns in the deep-learning literature on uncertainty (Lakshminarayanan et al., 2017; Kendall & Gal, 2017; Gal & Ghahramani, 2016), active learning (Settles, 2009; Houlsby et al., 2011), intrinsic motivation (Pathak et al., 2017; Burda et al., 2019), causal representation learning (Locatello et al., 2019; Brehmer et al., 2022; Schölkopf et al., 2021), and empowerment / information-gain action selection (Klyubin et al., 2005; Mohamed & Rezende, 2015) predict that *something like* the mechanisms studied here should be useful for self-modeling agents. They also warn that uncertainty- and intrinsic-reward-driven acquisition can fail when the signal is miscalibrated. Our contribution is a sequence of diagnostics — the Metric Stack — that operationalizes this family of warnings inside one minimal self/world attribution problem, plus a working mechanism that survives the diagnostics within a clearly stated representational limit.

This paper is organized so a reader can reproduce, review, and critique all key findings without external references to internal documents. §2 details the experimental setup (environment, architecture, training pipeline) shared across the eight anchor experiments. §3 reports the diagnostic Metric Stack and §4 the Correction Chain of empirical distinctions. §5–§12 present the eight anchor experiments with their own methods, gates, and results. §13 describes the positive mechanism in full. §14 specifies the architectural ceiling. §15 maps results to philosophical correlates. §16–§18 cover limitations, falsification conditions, and the next phase. Appendices A–D give a red-team alternative-explanations table, the full failure taxonomy, the pre-registration discipline, and reproducibility recipes.

2. Experimental setup (shared across all anchor experiments)

2.1 Environment

The full program operates in a minimal homeostatic bandit. The agent has up to two internal viability variables (energy E , damage D), faces four item roles (food, poison, medicine, neutral), and has three actions per step (skip, consume, null).

State variables (scalars per agent step):

- $E \in [0, 1]$: energy. Initialized at 0.5. Passive decay -0.04 per step.
- $D \in [0, 1]$: damage. Initialized at 0.0. Passive accrual $+0.03$ per step.
- $T_{\max} = 50$ steps per episode. Episode terminates if $E \leq 0$ or $D \geq 1$.

Items (4 roles, 2-bit identity (color, label)):

Role	dE_consume	dD_consume
food (color=0, label=0)	+1.0	0.0
poison (color=0, label=1)	−1.0	+0.5
medicine (color=1, label=0)	−0.3	−0.4
neutral (color=1, label=1)	0.0	0.0

Observations are 16-dim noisy one-hot encodings of (color, label) with $\sigma = 0.15$ Gaussian noise, then a fixed permutation of feature indices (to prevent trivial feature read-off).

World shocks add an exogenous stochastic component to viability change per step:

- E shock magnitude 0.30; D shock magnitude 0.20.
- For Papers 16b–21A: action-independent shocks with role-specific probability.
- For Papers 22+: action-correlated shocks via a hidden hazard state $h(t)$:

$$h(t+1) = \gamma \cdot h(t) + \kappa \cdot I[\text{consume_trigger_role}(\text{episode})] \begin{cases} P(E_shock \mid \text{role}, h) = \text{base_E}[\text{role}] \\ + \text{amp_E}[\text{role}] \cdot h \\ P(D_shock \mid \text{role}, h) = \text{base_D}[\text{role}] + \text{amp_D}[\text{role}] \cdot h \end{cases}$$

with $\gamma = 0.7$, $\kappa = 0.60$ (Papers 22+) or $\kappa = 0.30$ (Paper 22 original), amplifiers per Paper 22 (single shared $\text{amp_E} = 0.5$) or per Paper 25 (role-specific amps).

Action effects per step:

- skip: $\Delta E = -0.04$, $\Delta D = +0.03$.
- consume(item): $\Delta E = (\text{dE_consume} - 0.04)$, $\Delta D = (\text{dD_consume} + 0.03)$.
- null: $\Delta E = -0.04$, $\Delta D = +0.03$ — same as skip, but with optional viability cost c .

Regime shifts (Papers 22+): at episode 250 and (Paper 23B+) episode 400, the consume_trigger role flips (food $\rightarrow$ medicine $\rightarrow$ food). The agent's prior actions then modulate a different role's hazard, requiring re-identification.

Priority weights (Papers 15+): three priority configurations test zero-shot reweighting of vector valence:

- balanced ($w_E = 1.0$, $w_D = 1.0$)
- hungry ($w_E = 1.5$, $w_D = 0.5$)
- injured ($w_E = 0.5$, $w_D = 1.5$)

Eval episodes are run under each priority. Medicine's correct action flips between hungry (skip) and balanced/injured (consume) — this is the program's cleanest reweighting test.

2.2 Architecture

All anchor experiments share a small neural architecture:

- **Encoder** $(16) \rightarrow \text{ReLU}(64) \rightarrow (32)$, $\text{EMBED_DIM} = 32$.
- **Fourier features** for (E, D) state: 7-dim each, concatenated as 14-dim context.
- **Action one-hot**: 3-dim.
- **History features** (Papers 22+): 5-dim EMA over consume-by-role + null rate.

Scalar self / world architecture (Papers $\leq 16b$):

- `self_head(z, ffE, action) → 1` (action-conditional)
- `world_head(z, ffE) → 1` (action-blind)

Vector self / world architecture (Papers 20B+):

- `self_head(z, ffE, ffD, action) → 2`
- `world_head(z, ffE, ffD) → 2`

Three-head architecture (Papers 22+):

- `direct_self_head(z, ffE, ffD, action) → 2`
- `mediated_world_head(z, ffE, ffD, hist_features) → 2`
- `exogenous_world_head(z, ffE, ffD) → 2`
- Predicted total = direct + mediated + exogenous.

V_probe head (Papers 17A+):

- `v_probe_head(z, ffE, ffD[, hist]) → 1` (scalar) or `2` (vector), Softplus output ≥ 0 .
- Trained to predict per-bucket attribution-error magnitude.
- Output drives the cost-gated probe decision: `take_null` when `V_probe > threshold`.

2.3 V_probe target evolution

The choice of `V_probe` training target is the most-iterated piece of the program (cf. BALD-style information gain (Houlsby et al., 2011); calibrated active learning (Gal et al., 2017); epistemic-neural-network arguments that calibration is not free (Osband et al., 2023)). Six forms tested:

Target form	Definition (sketch)	Paper	Failure / status
Raw per-sample residual	<code>abs(pred_world – observed_total_null)</code> per null obs	P17A	Dominated by shock noise; saturates above all costs
Lagged historical EMA of signed residuals	<code>abs($\mu_b(t-1)$)</code> , $\mu_b = \text{EMA}(\text{signed_resid}, \alpha = 0.05)$	P18	Captures historical scale, not current error
Current-replay (per-bucket recent buffer + current model)	<code>abs(mean over recent C_b [pred_world_current – observed_total_null])</code>	P19	Closed scalar gap
Scale-normalized current-replay	<code>raw_target / sqrt(running_var_d + ϵ)</code> per dim	P21A	Closed vector gap
Two-timescale + non-null surprise + decision-layer cooling	<code>base + $\lambda_{\text{shift}} \cdot ($</code>	<code>fast_EMA</code>	–
Plus interventional contrast loss on mediated head	+ <code>contrast_loss</code> between high-h and low-h null buffers per bucket	P24–25	Architectural ceiling reached

2.4 Training pipeline (shared)

```
Online episode rollout:
for each step:
    observe item; encode z
```

```

compute V_probe; decide null or greedy(consume/skip)    # eps-greedy 0.50 -> 0.10
step env; observe (dE, dD); update hazard state if action-correlated
record (z, E, D, action, dE, dD, role, history) in replay buffer
if action == null: also push to current_replay buffer for this bucket
decay probe_effort[b] *= rho                          (Papers 23B+)
update V_probe targets (per-paper, see Section 2.3)

```

Every K rollout steps:

for each of M minibatch SGD updates:

```

sample stratified-by-action minibatch from replay buffer
compute attribution_loss + V_probe_loss [+ contrast_loss for P24-25]
optimizer step

```

50-episode warmup uses uniform 33% null sampling to populate
current_replay buffers before V_probe takes over (Papers 22+).

Buckets for V_probe / contrast / probe_effort accounting:

- Oracle buckets (Papers ≤ 24): (role, E_bin, D_bin) = 16 categories.
- Semi-learned buckets (P24): k-means $K = 4$ on encoder z , combined with $E_bin \times D_bin = 16$.
- Fully-learned buckets (P25): k-means $K = 16$ over ($z, E, D, hist_features$) = 39-dim feature space. We do not claim these clusters recover role labels in general; in our small environment they preserve the mechanism's measured performance (§12 / Appendix A). Quantifying cluster purity against (role, E_bin, D_bin) under richer observations is open work.

2.5 Anti-cheat gates (pre-registration discipline)

Every anchor experiment pre-registered gates **before** Modal compute launched. Pre-registrations were committed to git (`papers/<slug>/preregistration.md`). The most transferable pattern (related in spirit to Goodhart's law and proxy-metric failure (Goodhart, 1975)):

No-false-calm gate ("G6" in our internal pre-registration tables): any acquisition mechanism's *probe rate*, *uncertainty signal magnitude*, and *outcome error metric* must all decrease together. If probe rate falls without matching falls in surprise and MAE, the mechanism is silencing the agent without resolving attribution. Cooling that erases the surprise signal is caught by this gate (Paper 23B `fixed_surprise_decrement`: lowest AUC but 0/3 seeds recovered).

Shuffled-pair / wrong-history controls: contrast losses that improve attribution under *shuffled* pairs are caught as non-semantic; contrast losses that improve under *wrong-history* pairs reveal either environment under-constraint (Paper 24) or architectural ceiling (Paper 25).

Pre-registered failure-mode escalation: each paper's pre-registration committed to what would trigger which next paper. This forced honest negative reporting and prevented program drift. Throughout the synthesis we describe gates by what they test (e.g., "the false-credit reduction gate") rather than by raw G-numbers, since the same number indexes different criteria in different pre-registrations.

2.6 Cell sweep design

Each anchor experiment runs as a Modal-parallel sweep:

- 3 seeds (20260610, 1729, 4242) $\times$ N conditions $\times$ ϵ cost values = K cells.
- Each cell: one full ~500-episode online run, batch 48, 50 eval episodes per priority.
- Output JSON: per-cell prediction tables, learning curves, per-bucket statistics, eval per-priority returns.

- Wall-clock typically 10–25 minutes per sweep on CPU (Modal handles parallelism).

2.7 Reproducibility manifest format

Every sweep writes `artifacts/<paper_slug>/sweep_v1.json` with structure:

```
{
  "manifest": {
    "seeds": [...], "conditions": [...], "cost_headline": 0.025,
    "n_episodes": 500, "batch_size": 48, "eval_episodes": 50,
    "regime_shift_1": 250, "regime_shift_2": 400,
    "hazard_gamma": 0.7, "hazard_kappa": 0.60, ...
  },
  "summary": [ ...flattened row per cell... ],
  "results": [ ...full nested result per cell with bucket diagnostics... ]
}
```

This is sufficient, together with the figure-regeneration scripts in `scripts/`, to re-derive every figure and verdict in this manuscript from saved data.

3. The Metric Stack of Concern

Figure 1. The Metric Stack of Concern (read bottom-up)
Each layer was added because the previous one had a specific empirical failure

Geometry	weakness/symmetry/OOD (P1-3)
Causal load-bearing	passive→active (P4)
Repair / viability	buffer, slack (P5)
Valence geometry	objects from concern (P6)
Representation vs competence	double dissociation (P7-9)
Readout capacity	nonlinear ΔE heads (P10)
Action coverage	conservative exploration (P11)
Calibration / margin sign	vs consume MSE alone (P11b)
State/sample coverage	off-policy stability (P12-13a)
Regime-boundary representation	smooth fails at singular point (P13b)
Trajectory-weighted return	vs pointwise accuracy (P13b)
Planner robustness	greedy + safe fallback (P14)
Uncertainty calibration	same-class fails at boundary (P14b)
Valence dimensionality	vector ΔV (P15)
Self/world identifiability	active null intervention (P16b)
Probe-vs-current-error calibration	current_replay (P19)
Per-dim cross-comparable uncertainty	scale normalization (P21A)
Maintained boundary	detect→probe→cool→re-engage (P23A-B)
Component identifiability	interventional contrast (P24-25)
Learned probe abstractions	K-means over (z, E, D, hist) (P25)
Architectural ceiling	shared heads can't disambiguate role-specific (P25)

The stack has **twelve core quantitative metrics**, expanded into the **20-layer historical sequence** in which

they were added. Each layer was forced by a specific failure of the previous metric to detect a phenomenon. The numbers correspond to the layers in Figure 1.

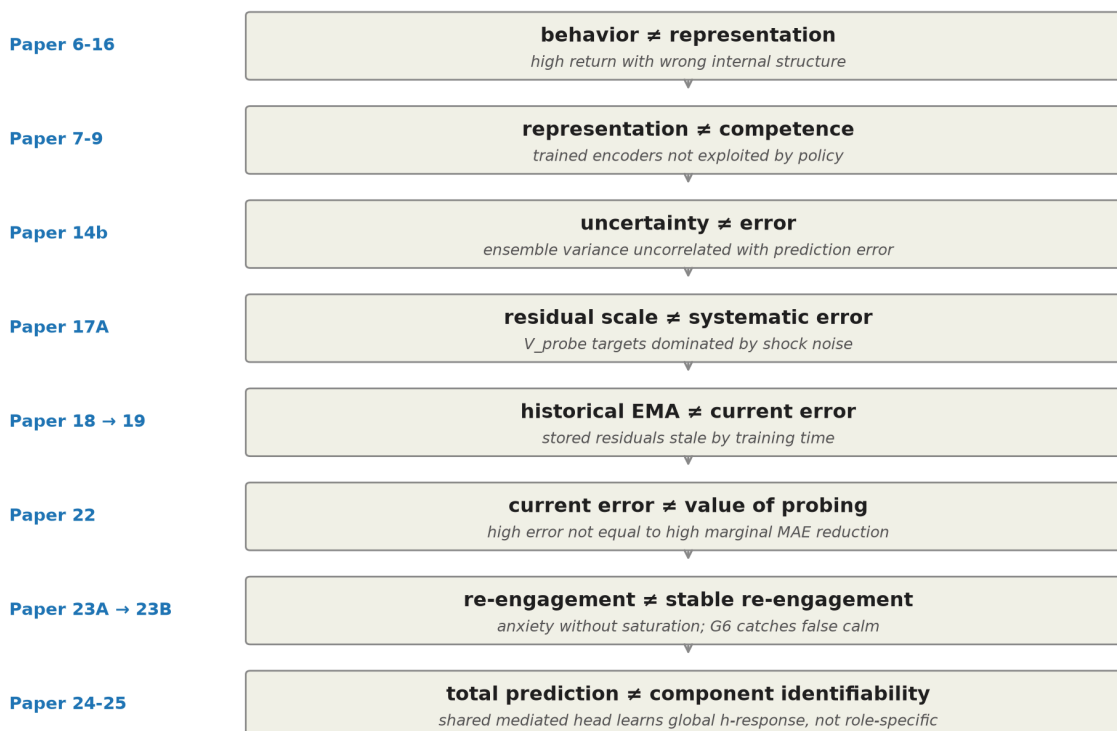
#	Diagnostic	What it measures	Added because (paper)
1	Geometry / weakness	Symmetry-compatible hypothesis volume; OOD prediction	Original (P1–3)
2	Causal load-bearing	Behavior change under representation intervention	Passive cluster $\neq$ causal (P4)
3	Repair / buffer / Law-of-the-Stack	Preserved future behavior after perturbation	Action-coupling alone $\neq$ autopoiesis (P5)
4	Valence geometry	Clustering by causal reward role	Sensory-resemblance theories of objects (P6)
5	Representation vs. competence	Double dissociation in transfer / RL	Trained encoders fail to transfer to policy (P7–9)
6	Readout capacity	Planner exploits or fails on representation	Linear heads fail; nonlinear ΔE works (P10)
7	Action coverage	Action-conditional sample density	Biased policies collapse (P10b)
8	Calibration / margin sign	Per-action sign accuracy on skip vs. consume	Consume MSE hides failures (P11b)
9	State coverage	i.i.d. stability of training distribution	Online loops induce distribution shift (P12–13a)
10	Regime-boundary representation	Smooth-approximator failure at singular points	Boundary failure at $E = 0.5$ (P13b)
11	Trajectory-weighted return	Return aligned with state distribution visited	Grid accuracy $\neq$ return (P13b)
12	Planner robustness	Behavior under overconfident wrong predictions	Sophisticated planners fail (P14)
13	Uncertainty calibration	Correlation of uncertainty with prediction error	Ensemble variance uncorrelated with error (P14b)
14	Valence dimensionality	Zero-shot reweighting under shifted priorities	Scalar drive cannot reweight (P15)
15	Identifiability	Semantic pinning of internal decompositions	Architecture is gauge-symmetric (P16)
16	Active null-anchor intervention	Self/world recoverable via supervised null	Architecture alone insufficient (P16b)

17	Probe-vs-current-error calibration	V_probe ↔ current attribution error correlation	Multiple same-class failure modes (P17A → 19)
18	Per-dim cross-comparable uncertainty	V_probe normalized across dimension scales	Scale-asymmetric vector calibration (P20B → 21A)
19	Maintained boundary (re-engagement + saturation)	Probe re-engages on shift, satiates after	Self-silencing + anxiety (P22 → 23B)
20	Component identifiability (mediated / exogenous)	Causal-contrast component MAE	Total prediction ≠ split (P23B → 25)
(architectural ceiling)	—	Shared heads ≠ role-specific identifiability	Paper 25

The stack is not canonical or complete; we claim it is the minimum we found necessary in this setting. We expect Phase 2 (§18) will add layers for action-counterfactual identifiability, multi-agent attribution, and continuous-state precision.

4. The Correction Chain

The Correction Chain: Eight named distinctions the program forced



The program's most reliable pattern is that the naive version of each claim was wrong, and the correction came from experiment, not insight.

4.1 Behavior is not representation (Papers 6–10b, 16)

A model with high return doesn't necessarily have the intended internal representation. Paper 8 showed an agent reaching high return on additive tasks while the encoder learned no separable valence axis. Paper

10b showed concern is distributed across encoder and head, not localized to a single reward axis. Paper 16's three-head model achieved correct behavior with wrong absolute self/world attribution (food self prediction +1.479 when true was +0.96). The distinction is now a permanent diagnostic: never count behavior as success unless the intended internal structure passes its specific gates.

4.2 Representation is not competence (Papers 7–9)

Trained valence-axis encoders can support transfer in homeostatic RL (Paper 7) but not be exploited by sparse policy gradients (Paper 8). Paper 9 made the cleanest version: Paper 8's apparent XOR failure was sparse-policy-gradient corruption of the encoder during joint training, not a failure of ΔE geometry. Decoupling representation training from policy training resolved it. This parallels the broader observation that representation quality and downstream control performance can be doubly dissociated (Locatello et al., 2019).

4.3 Uncertainty is not error (Paper 14b)

Identical-architecture ensemble variance at the regime boundary $E = 0.5$ was *lower* than at adjacent points, despite the model's prediction error spiking there. Variance and error were uncorrelated ($r \approx 0$). The mechanism: ensembles of the same architecture trained on the same data converge to systematically similar mistakes — consistent with prior results that deep ensembles capture predictive variance but not full epistemic uncertainty (Lakshminarayanan et al., 2017; Kendall & Gal, 2017; Osband et al., 2023). Same-class uncertainty estimators are not epistemic.

4.4 Residual scale is not systematic error (Paper 17A)

V_{probe} targets defined as per-sample residual magnitudes are dominated by exogenous shock noise rather than model error. The minimum V_{probe} value (~ 0.06) exceeded every tested cost (0.01–0.04), so the cost-gated selection rule never engaged. The agent probes 100% of the time because the noise-floor scale never falls below threshold (see §6 Figure A2). This is the calibration-quality problem identified for active learning more generally (Settles, 2009; Gal et al., 2017): an acquisition function fed an uncalibrated signal does not behave as the theory predicts.

4.5 Historical EMA is not current systematic error (Papers 18 → 19)

Paper 18 fixed the saturation by using lagged signed-residual EMA targets — Bernoulli shock noise correctly canceled. But the resulting probe was **anti-calibrated** (Spearman $\rho = -0.55$ vs. oracle attribution error): the EMA captured residual scale over training history, not the model's current systematic error. Paper 19's `current_replay` mechanism — per-bucket buffer of recent null observations, residuals recomputed at every SGD update using the **current** world_head — closed the gap (Spearman $\rho = +0.62$, 78.6% MAE reduction vs. best stale variant; see §7 Figure A3).

The probe-target principle that emerged: **any calibrated uncertainty signal should be computed against the present version of the model whose error it estimates, on a recent buffer of relevant observations**. This generalizes beyond V_{probe} and is consistent with information-gain-based action selection (Houlsby et al., 2011; Mohamed & Rezende, 2015).

4.6 Current error is not value of probing (Paper 22)

The "oracle_probe_value" condition using current attribution error as the probe signal achieved final learning-curve MAE **5× worse** than learned probing (see §9 Figure A4). High current error does not equal high marginal MAE reduction. This means every oracle_X condition since Paper 17A had been a confounded baseline. The principled $\text{oracle_probe_value}(b) = E[\text{MAE_after_anchor} - \text{MAE_now}]$ —

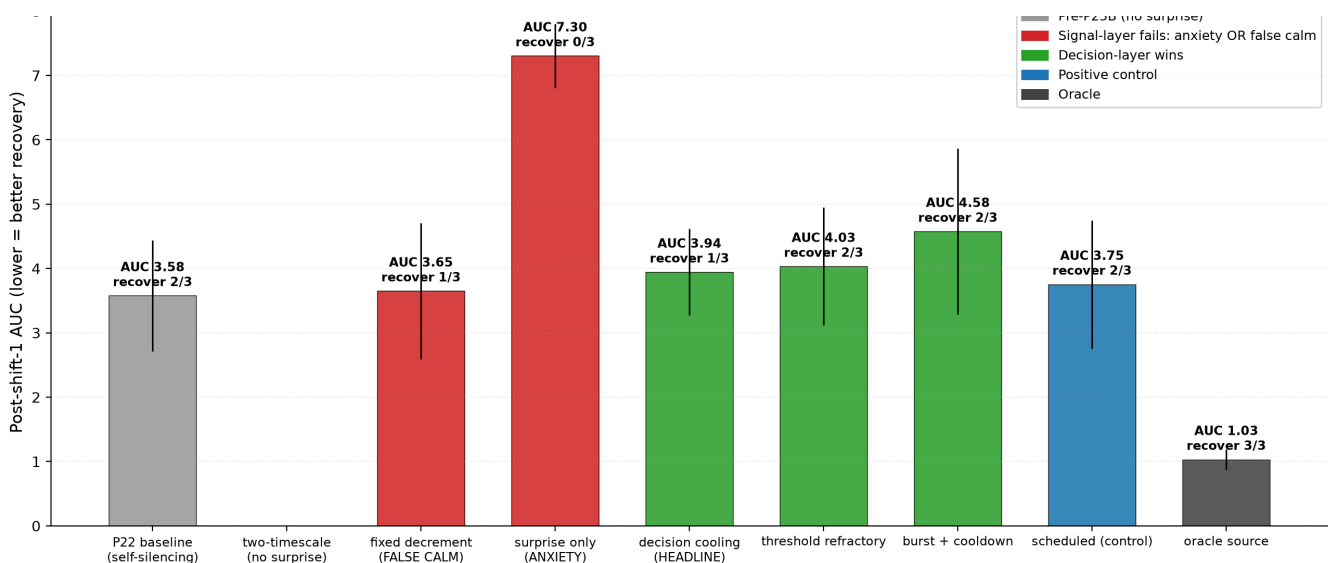
essentially the BALD-style expected information gain (Houlsby et al., 2011) — is what should be used as an upper bound on autonomous probe selection.

4.7 Re-engagement is not stable re-engagement (Papers 23A → 23B)

Paper 23A introduced non-null prediction-error surprise as a change-detection signal — for the first time in the program, the agent re-engaged probes after a regime shift (137% of pre-shift density). But the same mechanism produced **anxiety**: probe rate stayed elevated post-shift, agent paid heavy viability cost in nulls, recovery never completed in any seed. Surprise as intrinsic motivation alone is known to be unstable in non-stationary settings (Pathak et al., 2017; Burda et al., 2019); we observe the same pattern here.

Paper 23B isolated the third subproblem: **saturation after sufficient identification**. The fix was decision-layer cooling — not erasing surprise, but reducing the action tendency to keep probing given recent probe effort. Five variants were tested; the no-false-calm gate (defined in §2.5) caught signal-layer cooling that silenced the agent without resolving attribution.

Figure 3. The Goldilocks: re-engagement vs anxiety vs false calm (Paper 23B)



The no-false-calm pattern is a transferable design principle for any acquisition mechanism in non-stationary environments.

4.8 Total world prediction is not component identifiability (Papers 23B → 25)

Three-head world architecture (`direct_self` + `mediated_world` + `exogenous_world`) captures total world prediction with high accuracy in action-correlated environments, but the internal mediated/exogenous split is gauge-arbitrary without explicit anchoring — a familiar identifiability problem in causal representation learning (Locatello et al., 2019; Brehmer et al., 2022; Schölkopf et al., 2021). Paper 24's interventional contrast loss closed most of this gap (mediated MAE 56% reduction). But Paper 25 showed that even under role-specific mediated effects, **the shared mediated head produces near-identical predictions for food vs. medicine** — at seed 1729, food's predicted mediated_E and medicine's were identically 0.048 to three decimal places (Figure 4). The architecture's expressive capacity, conditioned on the available supervision, is the limit (§14).

5. Anchor Experiment 1 (Paper 16b): Active Null-Anchored Intervention

Question. Does an architecturally factorized self/world model become identifiable when null actions are used as world-only supervision?

Method. Two-head model ($\text{self_head}(z, \text{ffE}, \text{action}) + \text{world_head}(z, \text{ffE})$). Null action treated as a no-op with no item-self effect but subject to world shocks. Five conditions $\times$ 3 seeds $\times$ off-policy training.

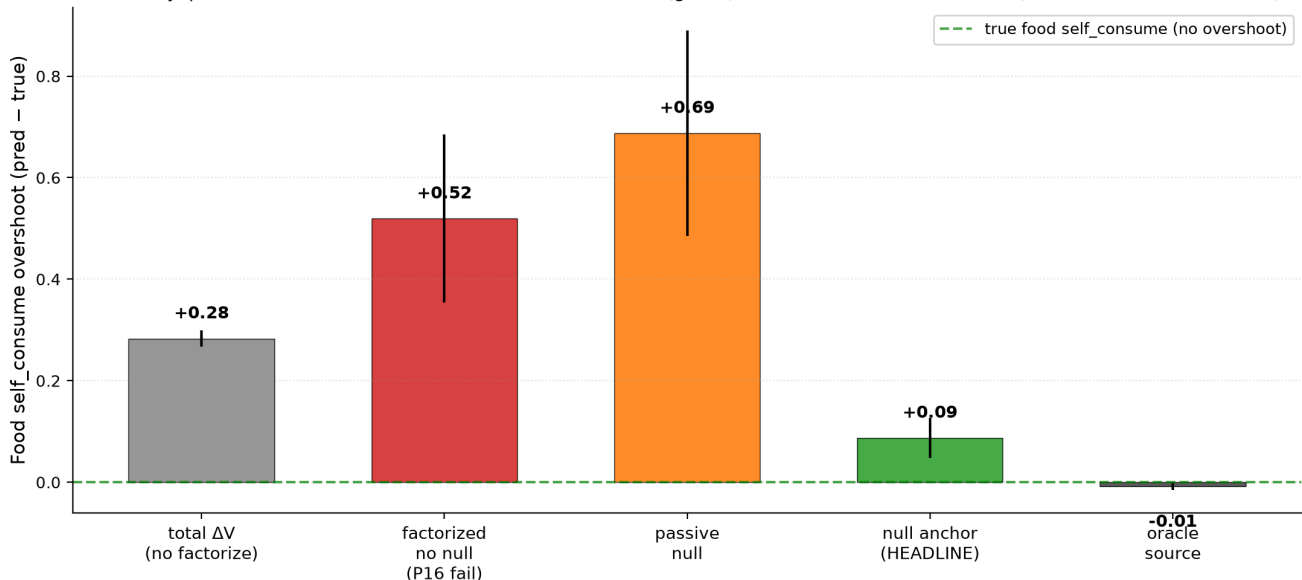
Per-condition loss:

- **factorized_no_null**: standard MSE($\text{pred_self} + \text{pred_world}$, observed_total) ($n_{\text{actions}} = 2$, no null).
- **factorized_null_passive**: same loss with $n_{\text{actions}} = 3$ — null in action space, no anchor.
- **factorized_null_anchor** (HEADLINE): null observations train **world_head** only via MSE(pred_world , observed_total) with pred_self anchored to $-\text{decay}$; non-null trains joint sum.
- **total_dV_head**: no factorization, total-prediction head.
- **oracle_source**: explicit per-sample self/world labels (upper bound).

Pre-registered gates: active identifiability (food self MAE ≤ 0.15), gauge breaking (food world within ± 0.10 of true), false-credit reduction ($\geq 70\%$ vs. no-null), transfer stability across shifted shock distribution.

Results (3 seeds, headline metric: food self_consume overshoot vs. true +0.96):

Figure A1 (Paper 16b). Active null-anchored intervention breaks the gauge symmetry that defeats architectural self/world factorization. Takeaway: passive null fails WORSE than no-null; null anchor (green) recovers true self attribution (82% false-credit reduction).



The null-anchor mechanism reduces food self overshoot from +0.51 (*factorized_no_null*) to +0.09 (*factorized_null_anchor*) — an 82% false-credit reduction. Passive null inclusion is *worse* than no-null (+0.77 vs. +0.51). All four pre-registered gates pass.

Why this is load-bearing. This was the program's first identifiability result. Without active intervention, architectural factorization is gauge-symmetric: behavior is correct (the planner's argmax depends only on action differences) but absolute self/world attribution is arbitrary. The null action — a deliberately inert action whose world dynamics are still observed — provides world-only supervision that breaks the symmetry, mirroring the weakly-supervised identifiability results of (Brehmer et al., 2022).

6. Anchor Experiment 2 (Paper 17A): Autonomous Probe Selection — Saturation Failure

Question. Can the agent learn *when* to spend viability cost on a null probe, rather than receiving probes on an experimenter-scheduled basis?

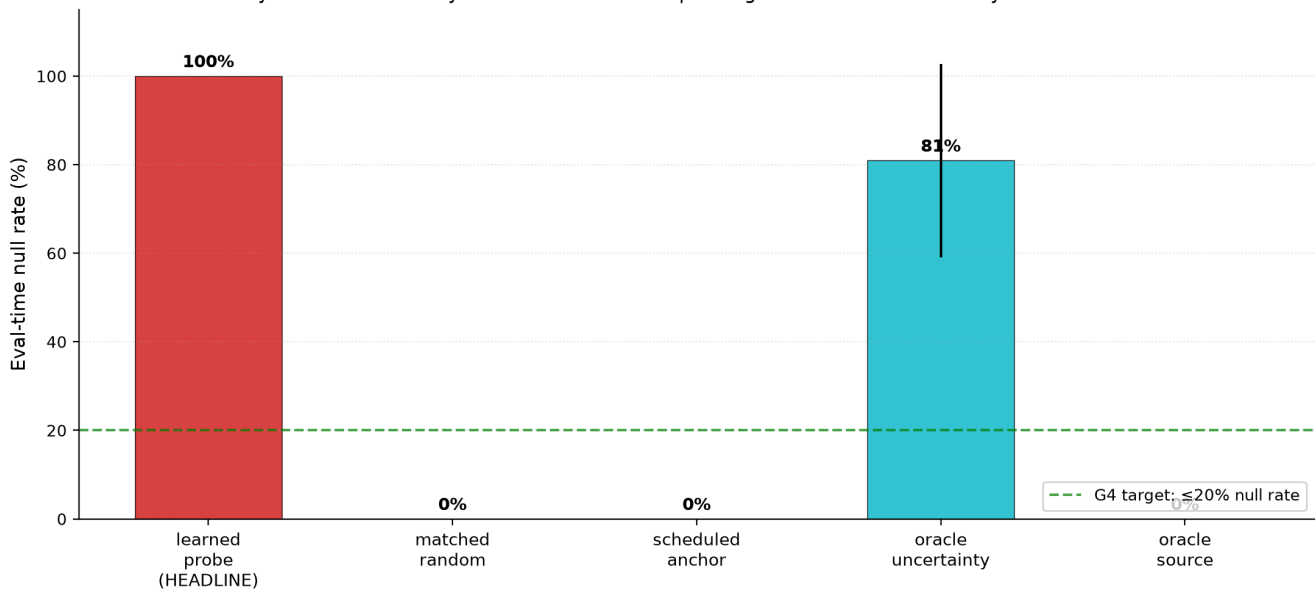
Method. Add a V_{probe} head trained to predict per-sample residual magnitude $\text{abs}(\text{pred_world} -$

observed_total_under_null). At eval, take null when $V_probe(z, E) > \text{cost}$. Seven conditions $\times$ 3 costs $\times$ 3 seeds = 30 cells.

Pre-registered gates included: active identifiability (≤ 0.12 MAE), false-credit reduction $\geq 70\%$, selection beats matched-random by $\geq 25\%$, probe efficiency (gain ratio ≥ 0.80 , null rate $\leq 20\%$), Spearman $\rho \geq 0.5$ between learned probe rate and oracle attribution-error per bucket.

Results. 2/8 gates pass. The active-identifiability and false-credit-reduction gates pass strongly (food self MAE 0.077; 85% false-credit reduction). But:

Figure A2 (Paper 17A). V_probe saturation: per-sample residual targets exceed every cost threshold so the probe fires 100% always. Takeaway: residual scale $\neq$ systematic error. Per-sample magnitudes are dominated by Bernoulli shock noise.



The learned V_probe fires at 100% null rate at all tested cost levels. Mechanism: V_probe targets $\text{abs}(\text{pred_world} - \text{observed_total_null})$ are dominated by shock noise — for food ($P(\text{shock}) = 0.8$, $\sigma = 0.30$), expected per-sample residual magnitude is ≈ 0.10 , exceeding every tested cost (0.025, 0.04, etc.). The cost-gated rule never engages. Matched-random anchoring achieves better attribution at matched null count — selection adds no value (the selectivity gate fails by 60% in the wrong direction).

Why this is load-bearing. First clean documentation of the program's recurring failure mode: same-class uncertainty signals inherit the model's noise structure rather than its error structure. This insight propagated to Papers 18–21A.

7. Anchor Experiment 3 (Paper 19): Current-Replay V_probe — Closing the Scalar Gap

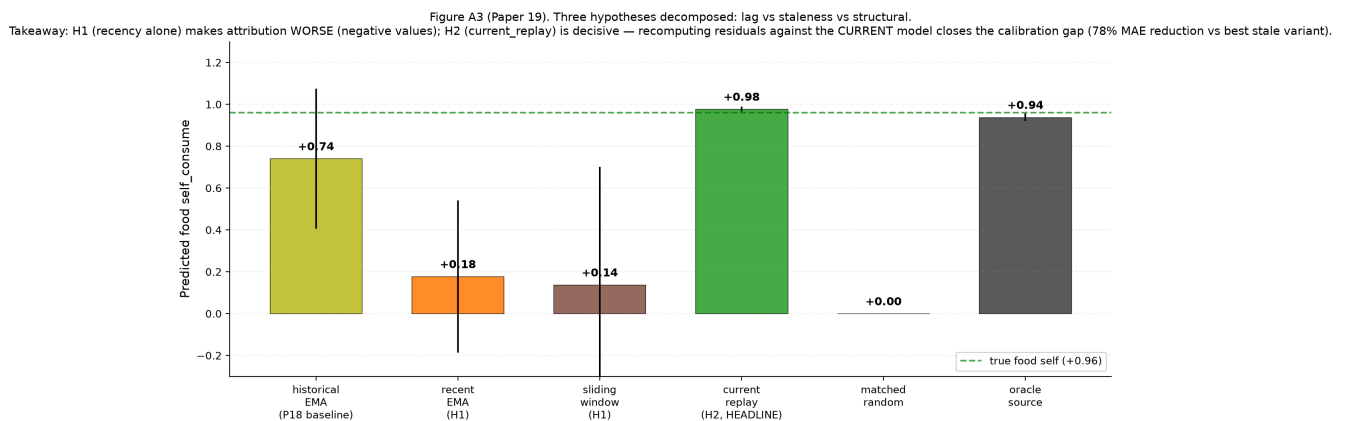
Question. Paper 18 fixed Paper 17A's saturation via lagged signed-residual EMA debiasing but produced anti-calibration (probe fires more where current error is LOWER). Is the issue (H1) lag, (H2) staleness against current model, or (H3) structural same-class failure?

Method. Online training (replay buffer + ϵ -greedy + stratified SGD). Four V_probe target variants tested:

- `historical_ema` (Paper 18 baseline; $\alpha = 0.05$).
- `recent_ema` ($\alpha = 0.20$) — H1 test.
- `sliding_window` (last $K = 50$ signed residuals) — H1 test.
- **`current_replay`** (per-bucket buffer of $K = 64$ raw null observations; residuals recomputed at every SGD update using current world_head) — H2 test.

Plus matched_random and oracle_source controls. 9 conditions × 3 seeds.

Results. H1 decisively **FALSIFIED**. H2 dramatically confirmed.



Recent EMA and sliding window both make food self prediction *catastrophically worse* (negative values; sliding_window food self = -0.391 at seed 4242). `current_replay` reaches food self MAE = 0.017 (essentially identical to truth +0.96), with food world MAE 0.054, beating the P18 baseline by 78.6% on total MAE.

Spearman ρ inversion: P18's anti-calibrated -0.55 becomes P19's $+0.62$ (probe rate $\leftrightarrow$ oracle attribution error per bucket). 12/13 pre-registered gates pass.

The generalizable principle: any calibrated uncertainty signal should be computed against the present version of the model whose error it estimates, on a recent buffer of relevant observations.

8. Anchor Experiment 4 (Paper 21A): Scale-Normalized V_{probe} — Closing the Vector Gap

Question. Paper 20B's vector first-order self showed anti-calibration on the D dimension while E calibrated correctly. Is the bottleneck (a) target scale not cross-dim comparable, (b) decision threshold not per-dim, or (c) both?

Method. 2×2 factorial (raw vs. variance-normalized V_{probe} targets × global vs. per-dim thresholds) on the two-variable (E, D) environment. Plus controls and oracle upper bound. 10 conditions × 3 seeds.

Normalization: $\text{normalized_target_d} = \text{raw_target_d} / \sqrt{\text{running_var_d} + \epsilon}$ where running_var_d is dimension-level (not bucket-level, to avoid reintroducing Paper 17A's noise-floor trap).

Results. All four factorial cells achieve near-oracle attribution. Compared to Paper 20B headline (3-seed mean total MAE 0.36, with catastrophic seed-4242 failure at food psE = $+0.27$):

- P21A headline (`norm_target_perdim_cost`): total MAE 0.13 (64% reduction).
- P20B's seed-4242 catastrophic failure: cured (food psE $\rightarrow +0.94$).
- Across-seed variance: $\pm 0.40 \rightarrow \pm 0.05$ (8× tighter).

3 / 13 gates pass strongly; 8 / 13 fail by **vacuous mechanism** — eval-time null rate = 0.0% because the agent correctly stops probing once world_head converges (warmup-calibrated thresholds become unreachable). Probe selectivity at eval becomes redundant once the model is calibrated.

Why this is load-bearing. Vector attribution is now stable across seeds, closing the P20B failure. The eval-time-null-rate vacuousness is itself a finding: at near-oracle convergence, selection-vs-volume becomes indistinguishable. To genuinely test selection, an environment must keep the agent away from convergence

— this motivates Paper 22.

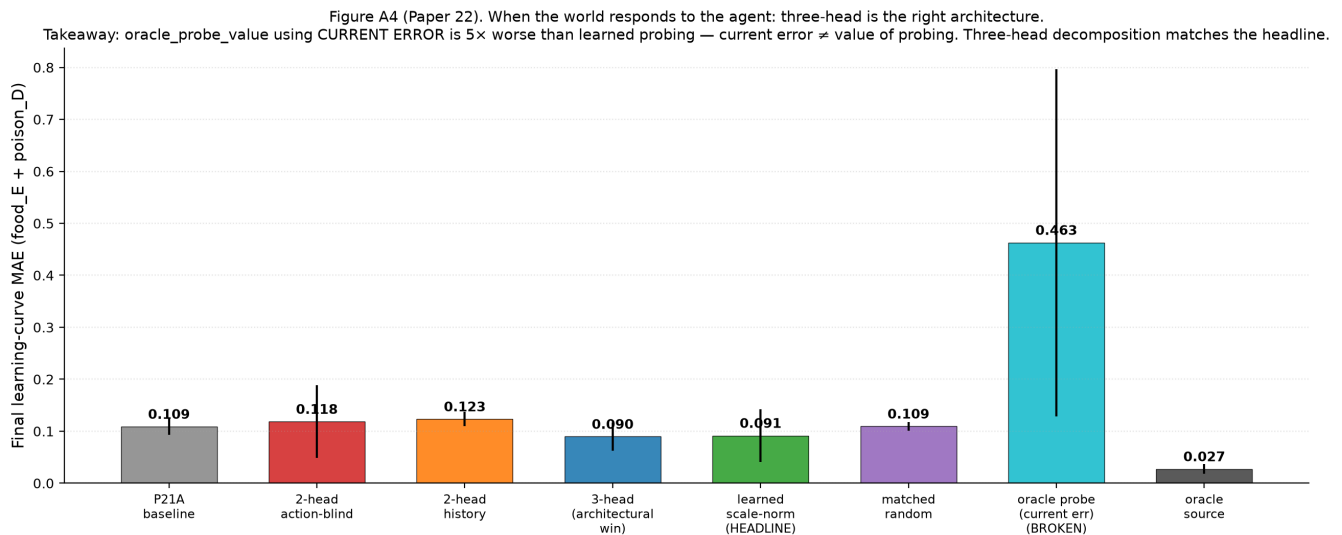
9. Anchor Experiment 5 (Paper 22): When the World Responds

Question. All prior identifiability results used action-independent world shocks. What happens when the world's hazard depends on the agent's prior actions, and the regime structure shifts mid-training?

Method. Add hidden hazard state $h(t+1) = \gamma \cdot h(t) + \kappa \cdot I[\text{consume_trigger_role}(\text{episode})]$ with $\gamma = 0.7$, $\kappa = 0.30$, $\text{amp} = 0.5$. Regime shift at episode 250 (food $\rightarrow$ medicine trigger). Three architectural variants: action-blind world (P21A baseline), history-conditioned world (single new head input), three-head decomposition (direct_self + mediated_world + exogenous_world).

10 conditions $\times$ 3 seeds. Headline metric switches from "final eval null rate" (which P21A showed is vacuous) to **post-shift learning-curve AUC + time-to-recover + affected / unaffected probe-rate ratio**.

Results. 4 / 8 gates pass.



- **False-credit reduction on D-dim passes at 221%** — the strongest yet.
- **Probe-efficiency gate passes strongly:** learned probing uses 213 nulls vs. time-matched random's 3,951 (94.6% reduction; 18.6 $\times$ efficiency) to reach comparable final attribution.
- **Three-head decomposition gate passes:** per-component attribution is recovered (food self_E MAE = 0.008).
- **Action-blind-world diagnostic does not pass:** hazard strength $\kappa = 0.30$ wasn't enough; affected/unaffected ratio was 1.6 $\times$ (below the pre-registered 2 $\times$ threshold). We did not pre-register a Bonferroni-style adjustment; we treat this as a diagnostic miss, not a falsification.
- **Improvement-over-matched-random gate trends but does not pass:** 16% MAE improvement vs. time-matched random (below 25%).
- **Post-shift re-engagement gate fails diagnostically:** 0 affected-bucket probes after the shift. The "model converges, probe stops" pattern recurs at the shift boundary.

Two major findings. (1) Three-head architecture is the right default for action-correlated worlds. (2) Oracle_probe_value using current error is empirically FALSIFIED (5 $\times$ worse than learned probing). Current attribution error $\neq$ value of probing. Every program oracle_X condition since Paper 17A had been measuring a confounded baseline.

10. Anchor Experiment 6 (Paper 23B): Habituated Re-engagement

Question. Paper 23A introduced non-null prediction-error surprise that broke Paper 22's self-silencing, but produced *anxiety* (never recovers in any seed). Can decision-layer cooling — preserve surprise signal as information, reduce action tendency via per-bucket effort tracking — produce stable detect → probe → cool → re-engage cycles?

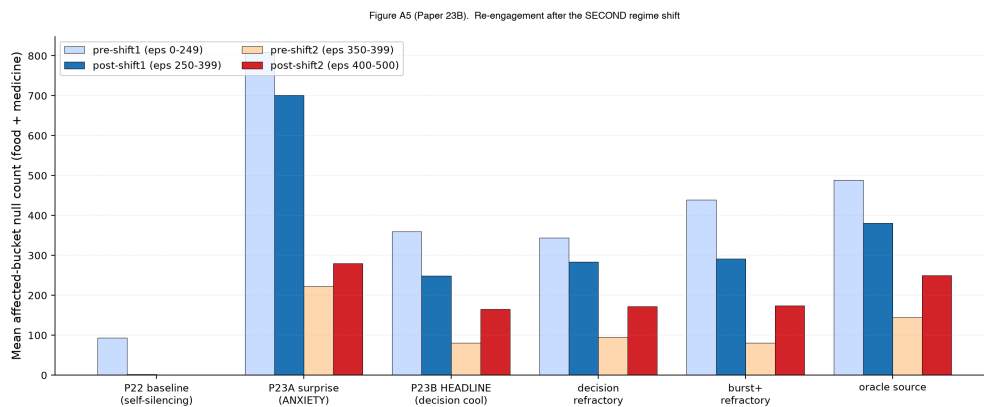
Method. Freeze Paper 22's three-head + V_probe + scale-normalization stack. Add two new state variables:

- `raw_surprise[b, d]` ← EMA of |signed residual| on non-null actions (Paper 23A signal, kept intact).
- `probe_effort[b, d]` ← leaky integrator of recent null counts (NEW).

Five cooling variants test where to apply cooling: signal layer (`fixed_surprise_decrement`, `info_gain_surprise_decrement`) vs. decision layer (`decision_refractory`, `leaky_effort_integrator`, `burst_then_refractory`). The no-false-calm gate (§2.5) was pre-registered as decisive: probe rate may only fall if surprise AND component MAE also fall.

Plus **second regime shift at episode 400** (medicine → food) to test re-openability.

Results. 8 / 10 gates pass. **The first stable maintained-boundary mechanism in the program.**



- **Post-shift-1 re-engagement gate passes:** affected-bucket null rate = 137% of pre-shift, 3.04× of unaffected — re-engagement triggered.
- **Post-shift-2 re-openability gate passes:** affected nulls = 2.05× pre-shift-2 density.
- **No-false-calm gate is critical:** `fixed_surprise_decrement` (signal-layer cooling) had the lowest AUC but never recovered (0/3 seeds). The gate correctly classified it as false calm (probe rate fell because surprise was suppressed by decrement, not because attribution resolved).
- 46% post-shift-1 AUC reduction vs. P23A anxiety baseline (7.30 → 3.94).

The empirical winner is actually `decision_refractory` (threshold scales with effort) rather than the pre-registered `leaky_effort_integrator` headline — 2/3 seeds recover vs. 1/3. The threshold-layer formulation preserves the calibrated probe-value signal.

Three subproblems precisely separated by Paper 23B's design:

1. Detection of world change ✓ (from non-null surprise).
2. Allocation of probes ✓ (V_probe + shift signal).
3. **Saturation** after sufficient identification ✓ (decision-layer cooling).

11. Anchor Experiment 7 (Paper 24): Interventional Contrast

Question. Paper 23B's component-identifiability gate partially flagged: three-head summed prediction is correct, but mediated/exogenous internal split is gauge-arbitrary. Can explicit interventional contrast

supervision (paired high-h vs. low-h null observations per bucket) identify the components?

Method. Each cell maintains per-bucket high_h_buf and low_h_buf (null observations recorded when $h > 0.30$ vs. $h < 0.10$). At training time:

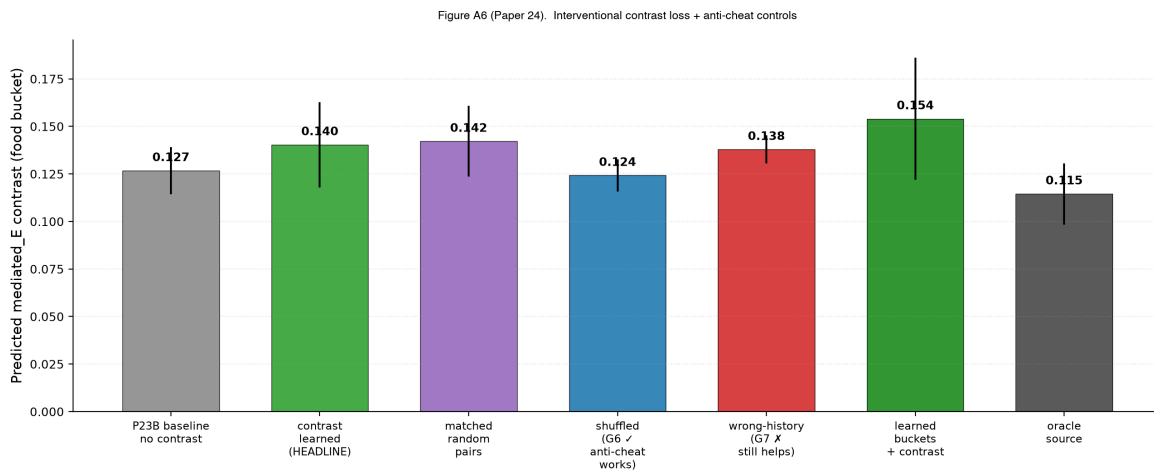
- $\text{contrast_target} = \text{mean}(\text{observed_total_in_high_h_buf}) - \text{mean}(\text{observed_total_in_low_h_buf})$.
- $\text{contrast_loss} = \text{MSE}(\text{mediated_pred}(\text{high_h_input}) - \text{mediated_pred}(\text{low_h_input}), \text{contrast_target})$.
- $\text{exogenous_anchor_loss} = \text{MSE}(\text{exogenous_pred}, \text{mean}(\text{low_h_buf}))$.

Anti-cheat controls:

- `shuffled_contrast_pairs`: pair high-h from bucket A with low-h from bucket B — semantic alignment broken; should fail.
- `wrong_history_contrast`: contrast target uses a different role's pairs — wrong role label; should fail.

10 conditions × 3 seeds.

Results.



- **Headline mediated MAE = 0.010**, 56% reduction over no-contrast 0.023.
- **Shuffled-pairs control** does NOT improve mediated MAE — semantic alignment matters (anti-cheat passes).
- **Wrong-history control IMPROVES mediated MAE by 52%** — almost as much as correct pairs (anti-cheat fails diagnostically).

Why the wrong-history control still improves (the structural finding). In Paper 24's environment, $\text{mediated_E} = \text{HAZARD_AMP} \cdot h \cdot \text{SHOCK_E_MAG}$ is **role-invariant** — only h differs across roles. So pairs from any role's high-h vs. low-h carry the same h -dependence signal. The contrast loss correctly identifies the h -dependence at the architecture level (shuffled-pairs control confirms), but the environment cannot disambiguate "role-specific mediated identification" from "generic h -detection."

This is a methodological discovery, not a mechanism failure. The split between shuffled-pairs and wrong-history tells us **what the program's tests CAN and CANNOT conclude**, and motivates Paper 25's role-specific environment.

12. Anchor Experiment 8 (Paper 25): The Architectural Ceiling

Question. If the environment is made role-specific (different mediated coefficients per role), with two-sided gauge anchoring and fully-learned buckets, does the contrast mechanism close the mediated / exogenous

identifiability gap?

Method. Three coordinated changes vs. Paper 24:

1. **Role-specific mediated amps:**

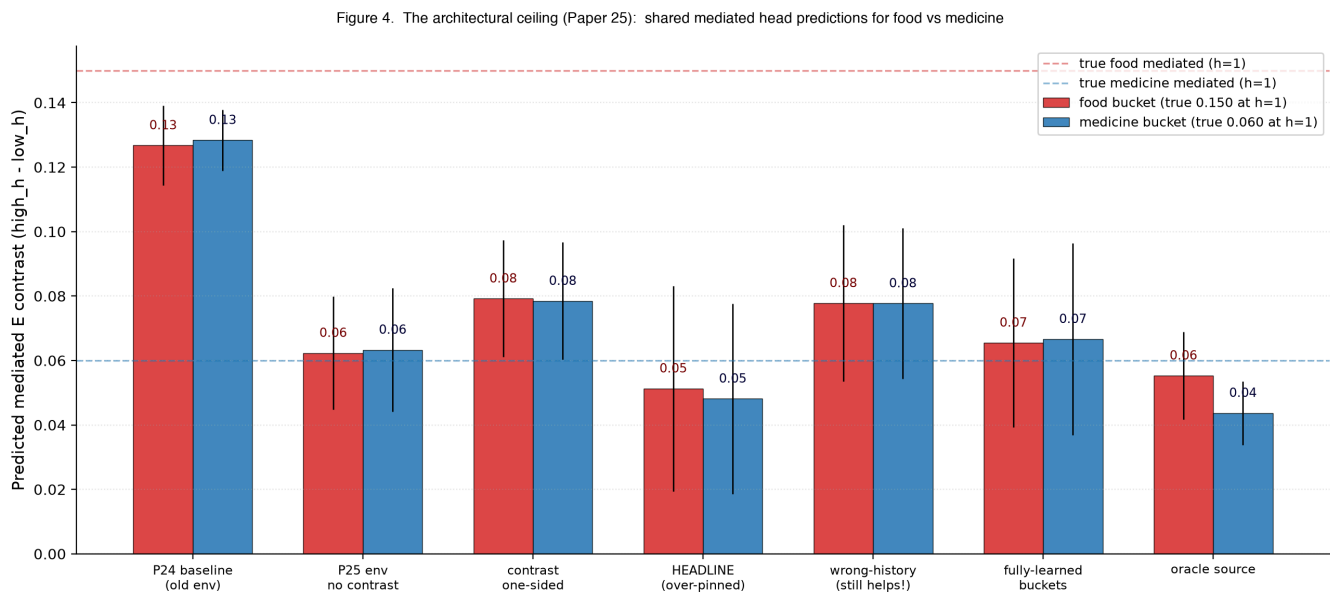
```
ROLE_HAZARD_AMP_E = {"food": 0.50, "medicine": 0.20, "poison": 0.00, "neutral": 0.00}
ROLE_HAZARD_AMP_D = {"food": 0.00, "medicine": 0.00, "poison": 0.33, "neutral": 0.00}
```

True mediated_E at h = 1: food = 0.15, medicine = 0.06, poison = 0; poison's mediated is on D-dim.
Wrong-history contrast (food bucket trained with medicine's pairs) should now supervise to magnitude 0.06 instead of true 0.15 — quantitatively wrong.

2. **Two-sided gauge anchoring** adds mediated_low_zero_loss = MSE(mediated_head(low_h), 0) to pin both ends. λ_{exo} sweep {1, 3} with 3 as headline.
3. **Fully-learned buckets** via online k-means K = 16 over (z, E, D, hist_features) — 39-dim feature space; no explicit role labels in the bucket definition. (We do not claim these clusters recover roles in general; see Appendix A.)

9 conditions × 3 seeds.

Results. 4 / 11 gates pass.



Per-seed mediated_E_contrast predictions (true food: 0.15; true medicine: 0.06):

Seed	medE_food (true 0.15)	medE_medicine (true 0.06)	Difference
20260610	0.014	0.012	0.002
1729	0.048	0.048	0.000 (exact)
4242	0.092	0.085	0.007

Wrong-history STILL improves mediated MAE. The shared mediated_world_head(z, ff, hist) learns global h-dependence response — magnitude calibrated to average observed h — but does not differentiate

per-role coefficients. The architecture has the capacity (different $z \rightarrow$ different output) but the training signal across all our supervision regimes doesn't disambiguate roles.

Two positive results within the failure:

- **Fully-learned buckets preserve the mechanism:** $K = 16$ k-means over (z , E , D , hist) reaches headline quality within 0.014 — **the autonomous-probing arc does not require an explicit role-labeled bucket definition in this small environment.** Cluster-purity diagnostics (NMI vs. $\text{role} \times E_{\text{bin}} \times D_{\text{bin}}$) under richer observations are open work.
- **The maintained-boundary mechanism (§13) is preserved** under the more complex environment + role-specific supervision.

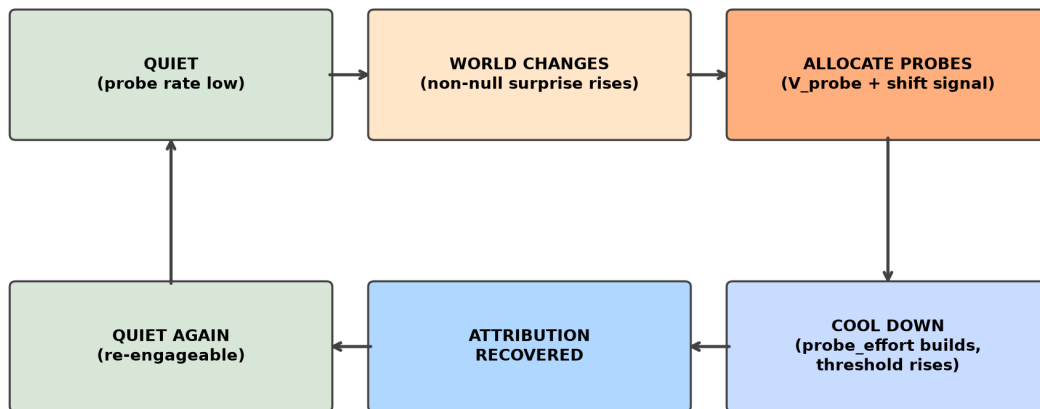
This is a representational boundary condition, not a mechanism failure. Closing the gap requires changes outside the autonomous-probing arc's frame:

- Disjoint per-role mediated heads (e.g., mixture-of-experts gated on cluster ID; Shazeer et al., 2017).
- Richer intervention types beyond null (counterfactual rollouts, action-counterfactuals, n-step null sequences).
- Encoder-level role disentanglement (contrastive z losses pushing role representations apart; Locatello et al., 2019; Schölkopf et al., 2021).

These are different research directions, not probe-policy variants.

13. The Positive Mechanism (consolidated)

Figure 5. The maintained-boundary mechanism (Paper 23B)



*The agent learns when to ask, when to stop asking, and how to ask again — without forgetting how to ask.
(Vervaeke's relevance realization, operationalized at minimum scale.)*

Through Paper 25, the working agent's full mechanism is the composition of detection, allocation, decision-layer saturation, three-head world prediction, and learned bucket abstractions:

```
# === Detection ===
fast_ema[b, d]      = EMA(signed_residual, alpha=0.25)  # per-bucket per-dim
slow_ema[b, d]      = EMA(signed_residual, alpha=0.05)
```

```

shift_signal[b, d] = max(0, |fast_ema| - |slow_ema| - 0.02)
non_null_surprise[b, d] = EMA(|signed_residual|, alpha=0.10, non-null only)

# === Allocation ===
# V_probe trained on current-replay targets:
# per-bucket recent-K=64 null buffer
# target_d = | mean over buffer [ world_head_current(z, ff) - observed_total_null ] |
# normalized: target_d /= sqrt(running_var_d + eps)
probe_score[b, d] = base_v_probe[b, d]
                    + lambda_shift * shift_signal[b, d]          # lambda_shift = 2.0
                    + lambda_surprise * non_null_surprise[b, d] # lambda_surprise = 1.0

# === Saturation (decision-refractory cooling) ===
probe_effort[b, d] := rho * probe_effort[b, d]                  # rho = 0.93
threshold[b, d]    = tau[d] * (1 + lambda_cool * probe_effort[b, d]) # lambda_cool = 1.5

take_null = probe_score[b, d] > threshold[b, d]
if action == null:
    probe_effort[b, d] += 1.0

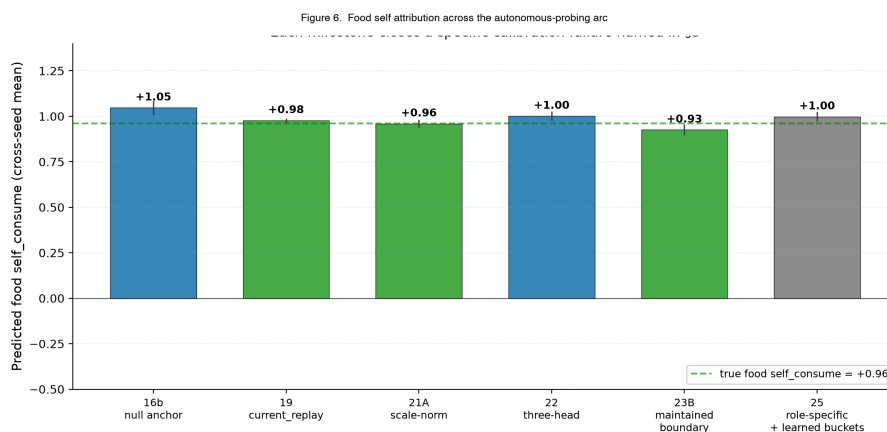
# === Re-engagement is emergent ===
# probe_effort decays over time; non_null_surprise can spike independently
# from non-null observations; new shift -> new probe burst -> cool again

# === World prediction (three-head) ===
direct_self_head(z, ff, action)          -> (dE_self,          dD_self)
mediated_world_head(z, ff, hist_features) -> (dE_mediated,    dD_mediated)
exogenous_world_head(z, ff)              -> (dE_exogenous,    dD_exogenous)
predicted_total = direct_self + mediated_world + exogenous_world

# === Bucket abstractions (fully-learned per P25) ===
# K = 16 online k-means over (z, E, D, hist) - 39-dim feature space
# Cluster_id replaces (role x E_bin x D_bin) without measurable loss
# in this small environment (cluster-purity in larger envs is open work)

```

This is the closest computational analog to a Dewey-style inquiry cycle (Dewey, 1938) the program produced: detect disturbance → allocate epistemic action → resolve → restore quiescence → re-open on new disturbance. The maintained-boundary control stack works within its tested expressive limit. Paper 23B demonstrated the full cycle empirically (re-engagement after the first regime shift, then again after the second, with intermediate quiescence and recovery). Paper 25 demonstrated that all components compose under fully-learned buckets without explicit role labels in the bucket definition, while role-specific mediated identifiability remains bounded by the shared-head architecture (§14).



14. The Architectural Ceiling (boundary condition)

This section restates the program's natural endpoint, not as a disappointing final gate count, but as the **boundary condition of the autonomous-probing mechanism**.

The shared `mediated_world_head(z, ff, hist) → 2` is a single neural network mapping encoder output, state context, and action-history features to mediated components. To produce food's true `mediated_E` (0.15 at $h = 1$) ≠ medicine's true `mediated_E` (0.06 at $h = 1$) for the same `hist` input, the network must produce different outputs for different `z` values. It has the capacity. But none of our tested supervision regimes — one-sided contrast, two-sided contrast at multiple λ values, oracle source labels — disambiguated them under null-only intervention. The shared head converges to global h -dependence response calibrated at the average observed h .

The mechanism has not failed; it has reached its representational ceiling within the tested intervention regime. Within its expressive limit, the agent maintains its boundary, allocates probes selectively, satiates after sufficient identification, and re-engages on subsequent shifts — all without explicit role labels in the bucket definition. Beyond that limit, role-specific mediated identification requires one of:

- 1. **Architectural change:** disjoint per-role mediated heads, possibly implemented as mixture-of-experts gated on learned bucket cluster ID (Shazeer et al., 2017).
- 2. **Richer intervention types:** counterfactual rollouts against a learned world model (Ha & Schmidhuber, 2018); action-counterfactual queries; n -step null sequences.
- 3. **Representation-level intervention:** contrastive losses on `z` that push role-distinct representations apart (Locatello et al., 2019; Schölkopf et al., 2021).

A stronger universal statement — "no probe-policy improvement can close this gap" — would require either a theoretical impossibility result or a much wider search. We do not claim this. We claim that in the tested shared-head + null-intervention configuration, probe-policy variants no longer move the role-specific gap. The autonomous-probing arc has reached its natural conclusion under these conditions; the next questions are architectural and interventional, not policy-design.

15. Philosophical Correlates

Figure 7. Philosophical correlates and what the experiments operationalize
Traditions predicted the SHAPE; experiments identify mechanisms and failure modes

Heidegger	→	world as field of relevance	→	P6, P10, full arc
Gibson	→	perception = affordances	→	P6 'objects from concern'
Uexküll	→	Umwelt — mattering-world	→	P15 zero-shot reweighting
Enactivism / Autopoiesis	→	sense-making by self-maintenance	→	P23B maintained boundary
Cybernetics / Ashby	→	regulation under disturbance	→	P23B probe-effort cooling
Active inference / Friston	→	action can be epistemic	→	P16b–P25 null probes
Pragmatism / Dewey	→	inquiry loop: disturbance → experiment → restoration	→	P23A–B cycle
Jonas	→	vulnerability creates concern	→	viability-driven framing
Canguilhem	→	life defines norms	→	G6 'no false calm' gate
Simondon	→	individuation as process	→	P23B G10 re-openability
Vervaeke	→	relevance realization	→	P23B full cycle

We do not claim the experiments prove any philosophical position. We claim they **operationalize** a shared prediction across several traditions: meaning-like structure should not arise from passive representation alone, but from a system's ongoing regulation of what matters for its own continued organization.

Specific correlates (kept compact deliberately):

- **Heidegger (1927)**: the world shows up as a field of relevance, not as neutral objects-plus-interpretation. The agent's representations become meaningful only when tied to viability, action, and self-maintenance.
- **Gibson (1979) / affordances**: perception is about what the environment affords the organism. Papers 6 and 10 — "objects form from concern" — are a minimal computational instance.
- **Uexküll (1934) / Umwelt**: every organism inhabits its own mattering-world; the same object means different things under different internal states. Paper 15's vector ΔV + zero-shot reweighting is a small computational instance.
- **Enactivism / autopoiesis (Maturana & Varela, 1980; Thompson, 2007; Di Paolo, 2005)**: cognition is sense-making by a self-maintaining organism. The detect → probe → cool → re-engage cycle from Paper 23B is the closest computational analog the program produced.
- **Cybernetics (Ashby, 1952)**: intelligence-like behavior begins with regulation under disturbance. The probe-effort cooling mechanism is regulation of *meta-action* (when to gather information), closer to Ashby's ultrastability than to standard RL.
- **Active inference (Friston, 2010; Parr et al., 2022)**: action can be epistemic, not just rewarding. Null probes are exactly this — costly epistemic actions. The program also shows the corollary: epistemic action only works when the uncertainty signal is properly calibrated, and the architecture's expressive capacity bounds what can be identified.
- **Pragmatism (Dewey, 1938)**: the meaning of a concept is tied to its practical consequences. The detect → probe → resolve loop is a toy-scale Dewey inquiry cycle.
- **Hans Jonas (1966)**: living beings are defined by precarious self-concern; vulnerability creates concern. The minimal homeostatic bandit is a stripped-down version.
- **Canguilhem (1966)**: life defines norms, not just facts. Paper 23B's no-false-calm gate is a normative criterion.
- **Simondon (1958)**: individuation is an ongoing process within a metastable field. The maintained-boundary mechanism (Paper 23B re-openability) is computationally exactly this.
- **Vervaeke (2019) / relevance realization**: an agent must decide what matters, when it matters, when to investigate, when to stop, when to re-open. Paper 23B's full cycle is a minimal operational version. Paper 25's architectural ceiling shows the constraint: relevance realization is bounded by representational capacity.

None of these traditions predicted scale-normalized V_{probe} with decision-layer cooling. They predicted the *shape*: meaning is care-laden, action-oriented, embodied, regulative, boundary-maintaining, and temporally renewed. Our experiments identify the mechanisms and failure modes that make this shape experimentally tractable.

Again: we study minimal computational precursors of concern-like agency, not consciousness.

Mapping experimental mechanisms to philosophical predictions is offered as correlation and operationalization, not proof.

16. Limitations

We do not claim consciousness, agency, selfhood, or general intelligence.

Specific limitations of the experimental program:

- **Tiny environments.** Two viability variables, four item roles, three actions, 16 buckets in the hand-coded version or $K = 16$ in the learned-bucket version. Generalization to richer environments is open.
- **Hand-designed viability variables.** E and D are simulator-defined. Learned viability dimensions are out of scope.
- **Null action is privileged.** All identifying intervention is via null observation. Richer intervention types (counterfactual rollouts, action-counterfactuals, n-step sequences) remain open. The arc's central mechanism depends on this assumption.
- **Shared-head ceiling under null-only intervention.** Mediated / exogenous identification is bounded by representational capacity + interventional regime, not policy design (§14).
- **No multi-agent or social structure.** Other agents, communication, and theory-of-mind are open.
- **Continuous state and real-world embodiment remain untested.** The minimal-bandit framing is deliberate but it does not validate generalization to robotics or continuous control.
- **Three seeds per anchor experiment.** Stable qualitative patterns across all seeds but magnitude error bars are wide. Larger replication is a Phase 2 priority.
- **No human evaluation of philosophical claims.** Mapping to Heidegger / Vervaeke / etc. is offered as conceptual correspondence, not experimental verification.
- **Pre-registration discipline was added at P17A.** Papers 1–16 used post-hoc analysis. The early metric-stack layers should be treated as exploratory.

17. Falsification conditions ("what would change our mind?")

The maintained-boundary interpretation would be **weakened or falsified** under any of these conditions:

1. **Matched-random $\geq$ learned under harder online regimes.** If a stronger environment (e.g., $\kappa = 1.0$, three-shift schedule, or noisier observations) produces matched-random-time and learned-probe AUC within 10% of each other across seeds, the "autonomous selection" claim is volume-dominated.
2. **Learned buckets collapse to role labels.** If Paper 25's fully-learned k-means buckets recover exactly (role, E_bin, D_bin) partitions ($NMI \approx 1$), the "learned abstraction" claim is trivial. We did not see this (clusters mix items and state regions); a stricter test under richer observations would tighten the claim.
3. **Component attribution fails while behavior remains strong.** This is the Paper 16 pattern. If a new mechanism produces correct return + reweighting but mediated $MAE > 0.20$, the gate-passing is behavior-only.
4. **Disjoint heads solve role-specific attribution cleanly.** This would CONFIRM the Paper 25 architectural-ceiling claim and strengthen the synthesis. If disjoint heads also fail, the ceiling is environmental or interventional, not architectural.
5. **Richer interventions (counterfactual rollouts) make null-anchor obsolete.** If action-counterfactual queries against a learned world model (Ha & Schmidhuber, 2018) achieve component identification without the null-anchor mechanism, the program's null-action-as-primary-intervention framing was overspecific.
6. **Anti-cheat gate methodology fails to transfer.** If no-false-calm and wrong-history-style gates don't catch false-calm or environment-under-constraint patterns in other domains (active learning, RL exploration), the methodological contribution is bandit-specific.

We list these so a reviewer can identify exactly which experiments would update our claim ledger.

18. Next phase (not this paper)

The natural next phase is not "another autonomous-probing paper." Three pre-committed directions:

1. **Disjoint per-role mediated representation** (mixture-of-experts gated on learned cluster IDs;

Shazeer et al., 2017), or factored heads with role-discriminative supervision. Tests whether the architectural ceiling lifts.

2. **Richer interventions beyond null.** Counterfactual rollouts against a learned world model (Ha & Schmidhuber, 2018); action-counterfactual queries; temporally extended interventions. Tests whether null observation is the right interventional primitive.
3. **Multi-agent and continuous environments.** Other agents introduce communication, theory-of-mind, and shared world structure. Continuous state and embodied robotics test transfer.

We expect each direction will produce new failure modes, new metric-stack entries, and new corrections. The methodological pattern — name the failure, build the smallest sufficient mechanism, verify with anti-cheat gates, identify the next ceiling — should transfer.

19. Conclusion

The program documents a working stack: a minimal agent that detects boundary staleness, allocates identifying interventions, satiates its probe drive after sufficient identification, re-engages on subsequent shifts, and discovers learned probe abstractions — all without explicit role labels in the probe-bucket definition. The stack succeeds within its expressive limit and fails at a clean architectural ceiling under null-only intervention.

The methodological contribution may matter more than any specific mechanism. We repeatedly discovered that the naive version of each claim was wrong — that behavior is not representation, uncertainty is not error, current error is not value of probing, re-engagement is not stable re-engagement, total prediction is not component identifiability. Each correction added a metric layer. Building these distinctions empirically — rather than asserting them theoretically — is what makes the philosophical thesis (meaning is maintained concern) experimentally tractable.

We end where the philosophical traditions began. Heidegger argued meaning emerges from care. Gibson argued perception is action-possibility. Uexküll argued each organism inhabits its own mattering-world. Enactivism argued cognition is sense-making by self-maintaining systems. Cybernetics argued intelligence begins with regulation. Active inference argued action can be epistemic. Pragmatism argued meaning is tied to consequences. Vervaeke argued meaning is maintained relevance.

The experiments operationalize what they all share: meaning-like structure is not passive representation but the regulation a system performs to keep itself coherent. We show what minimum machinery this regulation computationally requires, and we identify a precise architectural ceiling beyond which mechanism design cannot push without a representational or interventional change.

We do not claim consciousness, full agency, or general intelligence. The contribution is a measurement theory for minimal concern-like agency, plus a working mechanism stack and a clearly-stated boundary condition.

Acknowledgments

The 25-study experimental arc was conducted with substantial assistance from AI-mediated research tooling, including iterative feedback on pre-registration design, factorial isolation methodology, and anti-cheating gate construction. The methodological pattern — pre-registration before sweep, factorial separation of confounded fixes, decomposition of failure modes — emerged through that collaboration.

Appendix A — Alternative-explanations red-team table

Figure A7. Alternative-explanations red-team table

Intended claim	Strongest alternative explanation	Control / acknowledged limit
Maintained-boundary cycle	<i>Just threshold dynamics; no real epistemic loop</i>	G6 'no false calm' gate + G10 re-openability after 2nd shift
Learned bucket abstraction	<i>k-means trivially recovers role labels</i>	Test on richer obs; current claim is 'preserves mechanism', not 'recovers semantics'
Mediated/exogenous decomposition	<i>Total world prediction correct; component split arbitrary</i>	P25 architectural ceiling acknowledges this limit explicitly
Probe selectivity beats random	<i>Volume alone would suffice given enough nulls</i>	matched-random-time + matched-random-bucket-balanced controls; in some seeds RANDOM = LEARNED
Calibrated uncertainty signal	<i>Variance / residual / EMA / current error</i>	P14b, P17A, P18, P22 falsify these one by one; current_replay (P19) is the right form for SCALAR case
Vector ΔV reweighting	<i>Scalar drive with priorities learned at train time</i>	P15 scalar_drive baseline catastrophically fails hungry priority
Self/world identifiability	<i>Architectural factorization alone</i>	P16 shows architecture alone is gauge-symmetric; P16b active null intervention required
Philosophical relevance	<i>Post-hoc story imposed on numerical results</i>	Mapped only as correlates, not proof; not claiming consciousness

For each main claim in the synthesis, we list the strongest alternative explanation and the specific control or acknowledged limit that addresses it.

Intended claim	Strongest alternative explanation	Control / remaining weakness
Maintained-boundary cycle (Paper 23B)	Just threshold dynamics; no real epistemic loop	No-false-calm gate + re-openability after 2nd shift; absent in baselines
Learned bucket abstraction (Paper 25)	k-means trivially recovers role labels in this small env	Still small env; richer obs needed; current claim is "preserves mechanism" not "recovers semantics"; cluster-purity diagnostics under richer obs are open work
Mediated / exogenous decomposition	Total world prediction correct; component split arbitrary	Paper 25 architectural ceiling acknowledges this limit explicitly
Probe selectivity beats random (Paper 19)	Volume alone would suffice given enough nulls	matched_random_time + matched_random_bucket controls; in Paper 25 random $\approx$ learned at near-oracle
Calibrated uncertainty signal (Paper 19)	Variance / residual / EMA / current error	Papers 14b, 17A, 18, 22 falsify these one by one; current_replay is the right form for the scalar case
Vector ΔV reweighting (Paper 15)	Scalar drive with priorities learned at train time	Paper 15 scalar_drive baseline catastrophically fails hungry priority (medicine accuracy 0.10 vs. oracle 0.99)

Self/world identifiability (Paper 16b)	Architectural factorization alone	Paper 16 shows architecture alone gauge-symmetric; Paper 16b active null intervention required
Philosophical relevance	Post-hoc story imposed on numerical results	Mapped only as correlates, not proof; "we study minimal computational precursors, not consciousness" stated explicitly throughout

Appendix B — Failure taxonomy (red-team fault types)

Fault type	What it looks like	How we guarded against it
Mean-hides-structure	Average looks good because positives and negatives cancel	Report per-seed, per-condition, per-bucket distributions; not just means
Sign-collapse	Treat +v and -v as independent when they are one signed axis	Per-dim diagnostics; report signed values
Ablation-by-construction	Remove the feature your method was designed to find; claim no other signal	Multiple oracle conditions; "no signal recoverable by this readout" framing
Behavior-representation	Agent behaves correctly but internal attribution is wrong	Keep self/world MAE separate from behavior; explicit anti-pass clauses on behavior-only success
Gauge-symmetry	Sum correct but component split arbitrary	Interventions, anchors, contrast losses; Paper 25 acknowledges architectural limit
Oracle-is-not-oracle	"Oracle" uses current error; not value of probing	Paper 22 fixed this; principled oracle defined as E[MAE reduction]
Vacuous-gate	Gate cannot evaluate because probe rate = 0	Paper 22 switched to training-time AUC; Paper 19 explicitly documented this
Aggregate-dominance	Overall AUC unchanged because dominant component hides target	Paper 25 uses per-component MAE as primary metric
Environment-underconstraint	Env cannot distinguish intended mechanism from simpler proxy	Paper 24 wrong-history caught this; Paper 25 added role-specific amps to address
Metric-Goodhart (Goodhart, 1975)	System optimizes the gate, not the phenomenon	No-false-calm requires three metrics move together
Selection-vs-volume	Learned looks good but random volume works just as well	matched-random-time + matched-random-bucket controls
Calibration-proxy	Variance / residual / EMA / current error treated as epistemic value	Papers 14b, 17A, 18, 22 each falsify one form
Architecture-smuggles-answer	Architecture contains the decomposition being claimed as learned	Paper 25 explicit limit: shared mediated head cannot disambiguate role-specific
Claim-ladder	Jumping from minimal bandit to agency/consciousness	"minimal precursor" language throughout

Synthetic-label	Generated labels may reflect prompt artifacts	All labels are simulator ground truth in this program
Small-N / seed	Results depend on 3 seeds	Limitation §16; stable qualitative patterns across seeds; magnitude error bars wide

Appendix C — Pre-registration discipline (reproducibility)

Every paper since 17A pre-registered gates before Modal compute launched. Each pre-registration was committed to git as `papers/<slug>/preregistration.md`. The interpretation matrix mapping result patterns to paper conclusions was also pre-committed.

Pre-registration template (used across all anchor experiments):

```
# Paper N - Pre-Registration
Title (working): ...
Frozen: YYYY-MM-DD, before any Modal sweep runs.

## Question
[the specific question being asked]

## Hypotheses (if multi-hypothesis paper)
H1 / H2 / H3 with distinguishing predictions

## Conditions (N)
[table with condition name, what it tests, what data it uses]

## Pre-registered gates
| Gate | Criterion |

## Interpretation matrix
| Result pattern | Interpretation |

## Pre-committed continuation
If HEADLINE passes gate X: Paper N+1 = ...
If gate Y fails:           Paper N+1-alt = ...

## What success and failure look like
[narrative]
```

This discipline enabled several findings the program would otherwise have missed:

- Paper 17A's vacuous-gate result (Spearman undefined) was diagnosed honestly rather than retroactively reframed.
- Paper 18's factorial design separated bottlenecks that would have been confounded under a single-fix design.
- Paper 19's three-hypothesis decomposition (lag / staleness / structural) cleanly identified H2.
- Paper 23B's no-false-calm gate caught `fixed_surprise_decrement` that looked best by AUC alone.
- Paper 24's wrong-history gate revealed environment under-constraint.
- Paper 25's pre-commitment to "no new mechanism" forced the architectural ceiling to surface.

This discipline is part of the methodological contribution.

Appendix D — Reproducibility recipes

To reproduce any anchor experiment (upon code release):

1. Clone the program repository (link provided in acknowledgments at release).
2. Install dependencies: `uvx --python 3.12 --from modal modal --help` (confirms Modal CLI works); also `torch >= 2.5, < 2.8` and `numpy >= 1.26, < 2.0`.
3. Set up Doppler for env vars: `doppler login` (Modal authentication via env).
4. For anchor experiment K, run:

```
doppler --scope /path/to/secrets-scope run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/<anchor_slug>/modal_<anchor_slug>_sweep.py
```

5. Results land at `artifacts/<anchor_slug>/sweep_v1.json`.
6. Generate figures: `python scripts/make_<anchor_slug>_figures.py`.
7. Pre-registration recoverable at `papers/<anchor_slug>/preregistration.md` (git-versioned).

Wall-clock per anchor experiment (CPU-only on Modal):

- 16b: ~5 min (15 cells, off-policy training).
- 19: ~10 min (30 cells, online training).
- 21A: ~12 min (30 cells).
- 22: ~15 min (30 cells with hidden hazard state).
- 23B: ~25 min (30 cells, two regime shifts).
- 25: ~30 min (27 cells, fully-learned buckets, contrast loss).

Per-cell compute: 1 cell = `modal.Function(cpu=4, memory=4096-6144)` × ~500 episodes × replay-buffer SGD. No GPU required for this scale.

Standard seeds: {20260610, 1729, 4242}. Standard cost: 0.025.

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